

## 1. HOLOMORPHIC CONTINUATION

**Theorem 1** (Identity principle for holomorphic functions). *Let  $U \subset \mathbb{C}$  be a connected open set. Let  $f, g : U \rightarrow \mathbb{C}$  be holomorphic functions. If  $f = g$  on a set with a limit point in  $U$ , then  $f = g$  on all of  $U$ .*

**Corollary 2.** *Let  $U \subset \Omega \subseteq \mathbb{C}$  be open subsets, with  $U$  nonempty and  $\Omega$  connected. Let  $f : U \rightarrow \mathbb{C}$  be a holomorphic function. Then there is at most one extension of  $f$  to a holomorphic function  $\Omega \rightarrow \mathbb{C}$ .*

## 2. CAUCHY'S INTEGRAL FORMULA

**Theorem 3.** *Let  $f : U \rightarrow \mathbb{C}$  be a holomorphic function defined on an open subset  $U$ . Let  $\gamma$  be a closed rectifiable curve in  $U$ . Then  $\int_{\gamma} f(z) dz = 0$ .*

**Theorem 4.** *Let  $0 \leq a < b \leq \infty$ . Let  $f(z)$  be a holomorphic function on the annulus  $\{z \in \mathbb{C} : a < |z| < b\}$  given by a convergent Laurent series*

$$f(z) = \sum_{n \in \mathbb{Z}} c_n z^n.$$

(1) *For any  $r \in (a, b)$  and  $n \in \mathbb{Z}$ , we have*

$$\begin{aligned} c_n &= \oint_{|z|=r} \frac{f(z)}{z^n} \frac{dz}{2\pi i z} \\ &= \frac{1}{2\pi r^n} \int_{\theta=0}^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta. \end{aligned}$$

(2) *For each compact subset  $E$  of  $(a, b)$ , there exists  $M \geq 0$  so that for all  $r \in E$ , we have*

$$\sum_{n \in \mathbb{Z}} |c_n| r^n \leq M. \quad (1)$$

**Theorem 5.** *Let  $U$  be an open subset of  $\mathbb{C}$ , let  $f : U \rightarrow \mathbb{C}$  be meromorphic. Let  $\gamma$  be a smooth closed curve in  $U$ , oriented counterclockwise, that does not pass through any pole of  $f$ . Then*

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{\substack{z \in \text{interior}(\gamma) \\ \text{pole of } f}} \text{res}_z(f).$$

**Remark 6.** Let  $0 < r < R$ . Let  $f$  be a meromorphic function on a neighborhood of the annulus  $\{z : r < |z| < R\}$  that has no poles on either of the circles  $|z| = r, R$ . Then

$$\oint_{|z|=R} f(z) dz = \oint_{|z|=r} f(z) dz + 2\pi i \sum_{\substack{r < |z| < R \\ \text{pole of } f}} \text{res}_z(f).$$

## 3. HOLOMORPHY OF LIMITS AND SERIES

**Theorem 7.** *Let  $U$  be an open subset of the complex plane. Let  $f_n$  be a sequence of holomorphic functions on  $U$ .*

- (1) *Suppose that the sequence  $f_n$  converges pointwise to some function  $f$ , uniformly on compact subsets of  $U$ . Then  $f$  is holomorphic.*
- (2) *Suppose that the partial sums  $\sum_{n \leq N} f_n$  converge pointwise to some function  $f$ , uniformly on compact subsets of  $U$ . Then the sum  $\sum_n f_n$  is holomorphic.*

## 4. BLASHKE FACTORS

**Definition 8.** Let  $\alpha \in \mathbb{C}$  with  $|\alpha| < 1$ . The *Blashke factor*  $B_\alpha$  is the function

$$B_\alpha(z) := \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

*Lemma 9.* The Blashke factors enjoy the following properties:

- (1)  $B_\alpha(0) = \alpha$  and  $B_\alpha(\alpha) = 0$ .
- (2)  $B_\alpha$  defines a holomorphic automorphism of the unit disc  $\{z : |z| < 1\}$ , with inverse  $B_\alpha$  (i.e.,  $B_\alpha(B_\alpha(z)) = z$ ).
- (3)  $|B_\alpha(z)| = 1$  for all  $z$  with  $|z| = 1$ .

## 5. HARMONIC FUNCTIONS

Let  $U$  be an open subset of the plane  $\mathbb{C}$ .

## 5.1. Definition.

**Definition 10.** We say that a smooth function  $u : U \rightarrow \mathbb{R}$  is *harmonic* if  $u_{xx} + u_{yy} = 0$ .

## 5.2. Relation with holomorphic functions.

*Lemma 11.* Let  $f : U \rightarrow \mathbb{C}$  be holomorphic. Then, writing  $f = u + iv$ , with  $u, v : U \rightarrow \mathbb{R}$ , we have that  $u$  is harmonic. Conversely, if  $U$  is simply-connected, then every harmonic  $u$  arises in this way for some holomorphic  $f$ .

*Proof.* The first claim follows from the Cauchy-Riemann equations. For the second claim, we solve the Cauchy-Riemann equations to find  $v$  such that  $f = u + iv$  is holomorphic on each connected component of  $U$ .  $\square$

**5.3. Mean value theorems.** For the following results, let  $u : U \rightarrow \mathbb{R}$  be harmonic, and suppose that  $U$  contains the disc  $\{z : |z - z_0| \leq r\}$ .

*Lemma 12.* We have

$$u(z_0) = \int_0^{2\pi} u(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}.$$

*Proof.* We may assume that  $U$  is simply-connected. The claim then follows from the Cauchy integral formula applied to a holomorphic  $f$  with  $\Re(f) = u$ .  $\square$

*Lemma 13.* For  $|z - z_0| \leq r$ , we have

$$u(z) = \int_0^{2\pi} u(z_0 + re^{i\theta}) \Re \frac{re^{i\theta} + (z - z_0)}{re^{i\theta} - (z - z_0)} \frac{d\theta}{2\pi}.$$

*Proof.* We can reduce to the previous lemma using Blashke factors. Alternatively, we can reduce first to the case  $z_0 = 0$  and  $r = 1$  and  $0 < z < 1$ . Then Cauchy's integral formula reads

$$f(z) = \int_{w:|w|=r} f(w) \underbrace{\frac{1}{2} \left( \frac{w+z}{w-z} + \frac{w^{-1}+z}{w^{-1}-z} \right)}_{\Re\left(\frac{w+z}{w-z}\right)} \frac{dw}{2\pi i},$$

which yields the required formula upon taking real parts.  $\square$

*Lemma 14.* For  $|z - z_0| \leq r_0 < r$ , we have

$$|u(z)| \leq \frac{r + r_0}{r - r_0} \int_0^{2\pi} |u(z_0 + re^{i\theta})| \frac{d\theta}{2\pi}.$$

*Proof.* We apply Lemma 13 and majorize the integrand in absolute value.  $\square$

## 6. APPROXIMATE FACTORIZATIONS OF HOLOMORPHIC FUNCTIONS

The next lemma closely follows the presentation of Theorem 21 of <https://terrytao.wordpress.com/2014/12/05/supplement-2-a-little-bit-of-complex-and-fourier-analysis/>.

TODO: maybe revise to follow the presentation of [1, §3.9]

*Lemma 15.* Fix  $0 < c_2 < c_1 < 1$ . Let  $f$  be a holomorphic function, on a neighborhood of  $\bar{D}$ , where  $D := \{|z - z_0| < r\}$ . Assume that  $f(z_0) \neq 0$ . Assume given  $M \geq 1$  so that whenever  $|z - z_0| = r$ , we have

$$|f(z)| \leq M|f(z_0)|.$$

Let  $\rho$  run over the zeros of  $f$ , counted with multiplicity. Then

$$\#\{\rho : |\rho - z_0| \leq c_1 r\} \ll_{c_1} \log M$$

and

$$\frac{f'}{f}(z) = \sum_{|\rho - z_0| \leq c_1 r} \frac{1}{z - \rho} + O_{c_1, c_2} \left( \frac{\log M}{r} \right).$$

*Proof.* By replacing  $f(z)$  with  $f(z_0 + z)$ , we may assume that  $z_0 = 0$ . By writing  $f(z) = g(rz)$  (so that  $f'(z) = rg'(rz)$ , hence  $\frac{g'}{g}(rz) = \frac{1}{r} \frac{f'}{f}(z)$ ), we reduce to the case  $r = 1$ . Our task is then to show that

$$\#\{\rho : |\rho| \leq c_1\} \ll_{c_1} \log M \tag{2}$$

and that for  $|z| \leq c_2$ , we have

$$\frac{f'}{f}(z) = \sum_{|\rho| \leq c_1} \frac{1}{z - \rho} + O_{c_1, c_2}(\log M). \tag{3}$$

By Jensen's formula, we have

$$0 \leq \sum_{|\rho| \leq 1} \log \frac{1}{|\rho|} = \int_0^{2\pi} \frac{\log |f(e^{i\theta})|}{\log |f(0)|} \frac{d\theta}{2\pi} \leq \log M. \tag{4}$$

On the other hand, for  $|\rho| \leq c_1$ , we have  $\log 1/|\rho| \geq \log 1/c_1$ . It follows that

$$\#\{\rho : |\rho| \leq c_1\} \leq \frac{\log M}{\log 1/c_1} \ll_{c_1} \log M,$$

whence (2).

Turning to the proof of (3), we may write

$$f = g \prod_{|\rho| \leq 1} B_\rho,$$

where  $g$  is a holomorphic function on a neighborhood of  $\bar{D}$  that does not vanish on  $\bar{D}$ . We then have

$$\frac{f'}{f} = \frac{g'}{g} + \sum_{|\rho| \leq 1} \frac{B'_\rho}{B_\rho}.$$

Below, we estimate separately the contributions to (3) from  $g$ , from  $|\rho| \leq c_1$ , and from  $c_1 < |\rho| \leq 1$ . The result follows by combining these estimates.

We first estimate the contribution of  $g$ . We will show that

$$\frac{g'}{g}(z) = O_{c_1, c_2}(\log M), \quad (5)$$

We may normalize  $f$  so that  $g(0) = 1$ . Then, since  $g$  does not vanish, Jensen's formula reads

$$\int_0^{2\pi} \log|g(e^{i\theta})| \frac{d\theta}{2\pi} = 0. \quad (6)$$

On the other hand, our hypothesis reads

$$|g(e^{i\theta})| = |f(e^{i\theta})| \leq M|f(0)| = M \prod_{\rho} |\rho| \leq M,$$

hence

$$\log|g(e^{i\theta})| \leq \log M.$$

By combining this pointwise upper bound with the mean zero property (6), we deduce that

$$\int_0^{2\pi} |\log|g(e^{i\theta})|| \leq \frac{1}{2} \log M.$$

Since  $g$  does not vanish, we may find a holomorphic primitive  $G$  for  $g'/g$  (i.e.,  $G$  is a logarithm of  $g$ ). Lemma 14 gives, for  $|z| \leq c_2$ , the estimate

$$|\Re(G(z))| \ll_{c_2} \log M,$$

together with a similar bound for the derivatives of the real part of  $G$ . By the Cauchy–Riemann equations, the same bound holds for the first derivatives of the imaginary part of  $G$ , hence for those of  $G$  itself. The estimate (5) follows.

We next estimate the contribution from  $|\rho| \leq c_1$ . After the calculation

$$\frac{B'_\rho}{B_\rho}(z) = \frac{1}{z - \rho} - \frac{1}{z - 1/\bar{\rho}}, \quad (7)$$

one term of which matches up with a term on the right hand side of (3), we see that it suffices to show that

$$\sum_{|\rho| \leq c_1} \frac{1}{z - 1/\bar{\rho}} \ll_{c_1, c_2} \log M. \quad (8)$$

Since  $|z| \leq c_2$ , we have

$$\left| \frac{1}{z - 1/\bar{\rho}} \right| \leq \frac{1}{1/c_1 - c_2} \ll_{c_1, c_2} 1.$$

Therefore the required bound (8) follows from (2).

We turn finally to the contribution from  $c_1 < |\rho| \leq 1$ . For such an element  $\rho$ , and  $z$  with  $|z| \leq c_2$ , we have

$$z - \rho \asymp_{c_1, c_2} 1, \quad z - 1/\bar{\rho} \asymp_{c_1, c_2} 1,$$

where we recall that  $A \asymp B$  means that  $A \ll B \ll A$ . By cross-multiplying in (7), it follows that

$$\frac{B'_\rho}{B_\rho}(z) \asymp_{c_1, c_2} \rho - 1/\bar{\rho}.$$

On the other hand, using the Taylor expansion of the logarithm, we have

$$\rho - 1/\bar{\rho} \asymp_{c_1} 1 - |\rho|^2 \asymp_{c_2} \log \frac{1}{|\rho|}.$$

We thereby obtain

$$\sum_{c_1 < |\rho| \leq 1} \frac{B'_\rho}{B_\rho}(z) \ll_{c_1, c_2} \sum_{c_1 < |\rho| \leq 1} \log \frac{1}{|\rho|} \leq \log M, \quad (9)$$

where in the final step we appealed to (4).  $\square$

#### REFERENCES

- [1] E. C. Titchmarsh. *The theory of the Riemann zeta-function*. The Clarendon Press Oxford University Press, New York, second edition, 1986. Edited and with a preface by D. R. Heath-Brown.